

YERABATI VENKATA PRADYUMNA

+1 (979) 344 6695 | pradyumna1402@tamu.edu | [in Pradyumna Yerabati](#) | [pradyumnayerabati14](#)

College Station, Texas - 77840, USA

OBJECTIVE

I am a Master's thesis student with a strong interest in pursuing a career in academia, particularly in the area of algorithmic optimization and data-driven research. My academic training, including advanced coursework in Operating Systems, Analysis of Algorithms and Deep Learning, are shaping my interest in rigorous problem-solving, optimization, scalable data analysis. Through a Graduate Research Assistantship, I hope to further develop my research skills while contributing meaningfully to work that informs practice, policy, and the broader academic community.

EDUCATION

- Texas A & M University, College Station** 01/2025
Master of Science, Computer Science Texas, USA
- National Institute of Technology, Warangal** 06/2019 - 05/2023
Bachelor of Technology, Electrical and Electronics Engineering Warangal, India
 - GPA: 7.62/10.0
- SR Junior College** 06/2018
Intermediate Warangal, India
 - Grade: 97.9%
- Teja International High School** 06/2016
Secondary Education Warangal, India
 - GPA: 10/10

EXPERIENCE

- HCL Technologies** 07/2023 - Present
Technical Lead Bangalore, India
 - Led the optimization of the BigFix IVR backend infrastructure, reducing data processing time for over **50,000 client records** from **more than three days to just three hours**, significantly enhancing system scalability and operational efficiency.
 - Designed, developed, and deployed **robust mock server architectures** to simulate large-scale production environments, enabling comprehensive backend testing and accelerating feature development cycles.
 - Spearheaded the implementation of advanced techniques for **front-end API interception and UI testing**, improving front-end and back-end integration testing coverage and identifying nuanced performance bottlenecks early in the development process.
 - Built a fully functional **PDF Translator** by integrating **LangChain** with **LLama** models, enabling intelligent document parsing and language translation capabilities; enhanced the accuracy and efficiency of text extraction and multilingual support for real-world applications.
 - Leveraged strong foundations in **Data Structures and Algorithms** to optimize code performance and system architecture, leading to substantial improvements in backend stability and reliability.
 - Actively collaborated with cross-functional teams to architect solutions that met real-world enterprise demands, demonstrating leadership in both technical innovation and team coordination.
- Summer Internship** 05/2022 - 06/2022
Summer Trainee — Machine Learning and Deep Learning Warangal
 - Completed an intensive one-month training program focused on Machine Learning and Deep Learning techniques.
 - Gained hands-on experience with various machine learning algorithms for both regression and classification tasks.
 - Explored foundational and advanced concepts of Deep Neural Networks (DNNs), including Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs).
 - Developed and implemented multiple projects, including:
 - * Image Compression using Neural Networks
 - * Cancer Prediction using Classification Models
 - * Diabetes Prediction using Machine Learning Techniques

PROJECTS

Development of Simulated Server Environments

Designed and built multiple enterprise-grade server environments to simulate Tenable.io, Tenable.sc, and Rapid7 APIs, enabling backend performance testing, system validation, and real-world interaction simulation.

Frameworks/Libraries: Flask, Faker, Pandas, SQLite3

Tools and Technologies: Python, Postman, Jenkins, Git/GitHub, JMeter, MySQL, Excel

- **Tenable.io Mock Server (BigFix IVR 4.0)**
Designed a mock server to simulate Tenable.io server responses for load testing and backend benchmarking.
- **Tenable.sc Mock Server (BigFix IVR On-Prem)**
Built a customized mock environment to simulate Tenable.sc API workflows and improve query efficiency.
- **Rapid7 Mock Server**
Developed a mock server replicating Rapid7 API responses for backend testing and system validation.

BigFix IVR Backend Optimization

Designed and implemented optimization strategies to significantly reduce backend data processing time and enhance system performance.

- Reduced backend processing time for over 50,000 clients from three days to three hours, achieving substantial scalability improvements.
- Focused on backend performance tuning, load balancing, and real-time database management to ensure system reliability and efficiency.

BigFix IVR WebUI Automation

Built an automated testing framework to validate frontend-backend integration and capture real-time performance metrics.

Frameworks/Libraries: Node.js, JavaScript

Tools and Technologies: Puppeteer, Git/GitHub

- Developed a Puppeteer-based framework to intercept API response payloads and assess UI responsiveness under simulated user loads.
- Enabled early detection of performance bottlenecks and improved overall system stability during high-load conditions.

ACADEMIC PROJECTS

Distribution System Resilience Using Deep Reinforcement Learning

Final Year Project — Applied advanced deep reinforcement learning (DRL) techniques to enhance the resilience of power distribution systems under asynchronous and partially observable conditions.

- Designed and trained a deep neural network control agent using Q-learning and experience replay to autonomously learn optimal restoration policies in response to disturbances.
- Modeled dynamic power system scenarios on a modified IEEE 13-bus distribution system to evaluate DRL agent performance against traditional rule-based control strategies.
- Demonstrated the superior adaptability, learning efficiency, and decision-making capabilities of DRL under uncertain, asynchronous information settings.
- Integrated asynchronous communication, partial observability, and microgrid coordination challenges within the reinforcement learning framework.
- Explored real-world deployment challenges such as cybersecurity risks, communication latency, and the need for reliable sensor infrastructures.

Heart Disease Prediction Using Machine Learning Algorithms

Developed a classification-based machine learning model to predict heart disease using real-world medical datasets.

- Implemented and compared multiple ML algorithms, including Logistic Regression, K-Nearest Neighbors (K-NN), and Support Vector Machines (SVM).
- Conducted data preprocessing, feature selection, and hyperparameter tuning to optimize model performance.
- Evaluated models based on accuracy, precision, recall, and ROC-AUC metrics.

Food Recipe Application using MealDB API

Built a web application that retrieves food recipes and cooking instructions based on user-input ingredients, using external APIs.

- Integrated MealDB API to fetch dynamic recipe data based on user queries.
- Designed a responsive and user-friendly frontend interface for recipe search and display.
- Focused on API request handling, error management, and clean UI/UX design principles.

Face Detection using Haar Cascade

Developed a real-time face detection application leveraging the Haar Cascade object detection algorithm.

- Implemented face detection functionality using OpenCV and Haar Cascade Classifiers with live camera feed.
- Focused on feature extraction, object localization, and optimization of detection accuracy.
- Demonstrated practical understanding of computer vision concepts and real-world image processing techniques.

SKILLS

- **Programming Languages:** Python, Javascript, C++, SQL
- **Web Technologies:** HTML, CSS, JavaScript, Node.js, Express.js, Puppeteer, Postman
- **Database Systems:** MySQL, SQLite
- **Data Science & Machine Learning:** Supervised Learning (Classification and Regression), Deep Learning (ANNs, CNNs), Computer Vision (Haar Cascade, OpenCV), Data Preprocessing, Model Evaluation Metrics
- **DevOps & Version Control:** Git, GitHub, Jenkins(CI/CD)
- **Specialized Area:** Backend Development and System Optimization, Web Automation and Performance Validation, Machine Learning (Supervised Learning, Deep Learning Basics)
- **Tools & Technologies:** Flask, Faker, Pandas, NumPy, Scikit-learn, OpenCV, SQLite3

ACHIEVEMENTS

- Published a white paper titled *From 3 Days to 3 Hours* on IVR backend optimization in the HCL Central Engineering Newsletter.
- Designed a proof of concept (PoC) for the development of simulated server environments (Mock Servers) and API interception using Puppeteer.
- Received monetary rewards for exceptional performance and technical contributions within the team.
- Solved over 300 problems on LeetCode and GeeksforGeeks, demonstrating strong algorithmic and problem-solving skills.
- Achieved a Gold Badge and 5-Star Rating for SQL on HackerRank.
- Secured an **A grade** in the **Data Structures and Algorithms** course during the 3rd semester.
- Attained a **99.11 percentile** in JEE Mains 2019, ranking among the top candidates out of 0.96 million applicants.
- Secured third place in the district-level badminton tournament (Sub-Junior Category), demonstrating competitive spirit and athletic excellence.

LEADERSHIP EXPERIENCE

• Technical Lead

2023 - Present

HCLTech

- Led the end-to-end execution of technical projects, including backend optimization, simulated server development, and frontend automation testing.
- Proposed and designed proof-of-concept (PoC) architectures for simulated server environments and API interception frameworks, showcasing initiative and innovation.
- Took complete ownership of project delivery while maintaining strong collaboration and coordination with cross-functional teams.
- Demonstrated leadership by driving technical initiatives independently, ensuring timely completion and high-quality outcomes.
- Led technical research initiatives in the final-year BTech project on Deep Reinforcement Learning for distribution system resilience, handling experiment design, simulation execution, and reporting of findings.

VOLUNTEER EXPERIENCE

- **Volunteer**

2022 - 2025

GWMC Badminton Stadium

- Volunteered in organizing and conducting badminton tournaments for two consecutive years during the Shiva Ratri festival events.

CERTIFICATIONS

- **Python Programming A-Z: Beginner to Expert Course** — Completed from April 2021 to June 2021.
- **Machine Learning and Deep Learning Applications** — Completed from May 2022 to June 2022.
- **The Web Developer Bootcamp 2022** — Udemy (Completed in 2022).
- **Certified in JMeter for Performance Testing and Load Testing.**
- **Certified in BlazeMeter for Advanced Performance Testing and Test Automation.**

ADDITIONAL INFORMATION

Languages: English (Fluent), Telugu (Native), Hindi (Professional Working Proficiency)

Interests: Backend Development and System Optimization, Web Application Development, Data Science and Machine Learning, Artificial Intelligence, Badminton, Yoga and Mindfulness, Research and Innovation